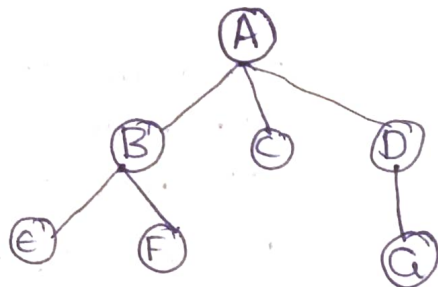


Activity - 31) Activity - 1 :- Tree Terminology

Root :- A

Parent of G :- D

Children of B :- E, F

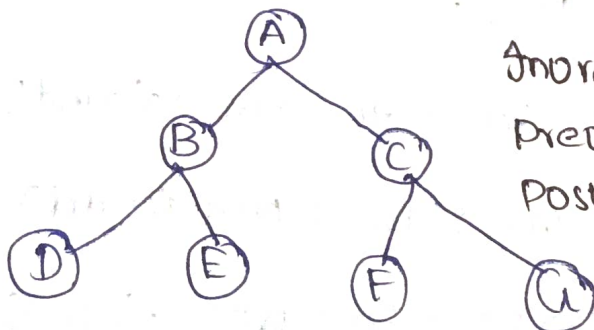
Leaf nodes :- C, E, F, G

Internal nodes :- A, B, D

Degree of B :- 2

Height of Tree :- 2

Level of F :- 2 (Root = level 0)

2) Activity - 2 :- Traversal Race

Inorder (LNR) :- D B E A F C G

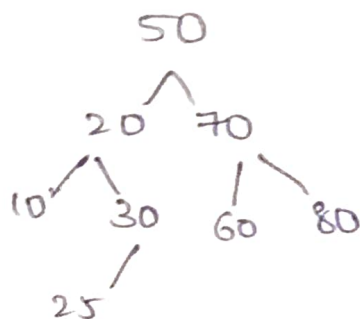
Preorder (NLR) :- A B D E C F G

Postorder (LRN) :- D E B F G C A

3) Activity - 3 :- BST Construction

Insert: 50, 20, 70, 10, 30, 60, 80, 25

BST:-



Minimum = 10

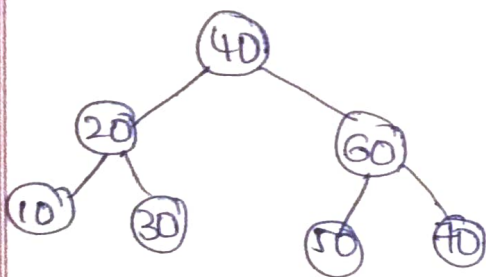
Maximum = 80

Height = 3

Leaf nodes = 10, 25, 60, 80

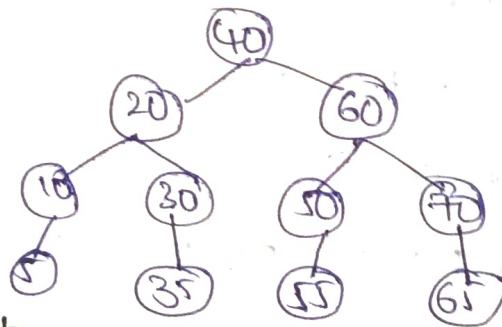
4) Activity - 4 :- BST Insertion

Initial BST:-



Insert: 55, 65, 5, 35

Updated BST:-



5) Activity - 5 :- BST Search

Using the updated BST:-

- Search 65 :- 40 → 60 → 70 → 65 (found)
- Search 90 :- 40 → 60 → 70 (Not found)
- Search 25 :- 40 → 20 → 30 (Not found)

6) Activity - 6 :- BST Deletion

→ Delete leaf node: 25

→ Delete node with one child: 50 (one child = 35)

→ Delete Node with two children: 20 (replace with inorder successor = 30)

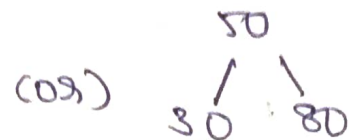
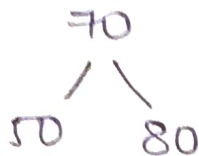
7) Activity - 7 :- Find the Mistake

Incorrect BST :-



Error:- left child (70) is greater than root (50).

Correct BST:-



either corrected BST follows BST rules.

8) Activity - 8 :- Quad Tree

100 x 100 Square:-

- (10, 20) → SW
- (80, 70) → NE
- (30, 60) → NW
- (90, 10) → SE

9) Activity - 9 :- Application Match

File System $\rightarrow$ Tree

Dictionary $\rightarrow$ BST

Google Maps $\rightarrow$ Quad Tree

3D Graphics $\rightarrow$ Octree

Game Collision $\rightarrow$ Octree

10) Activity - 10 :- Mini Hackathon

Example Numbers :- 50, 20, 70, 10, 30, 60, 80

Inorder :- 10, 20, 30, 50, 60, 70, 80

Preorder :- 50, 20, 10, 30, 70, 60, 80

Postorder :- 10, 30, 20, 60, 80, 70, 50

Minimum :- 10

Maximum :- 80

Delete Node :- 10

Case used :- Leaf Node Deletion.